PH621 - Computational Final Project
Exploring Realistic Systems in Classical Mechanics

October 9, 2026

Abstract

The objective of the computational final project is to investigate a realistic system in classical mechanics using numerical methods. Each project should apply concepts from the course to explore numerically the classical dynamics of a chosen physical system. This project is intended to serve as a bridge from textbook-level problems to the sorts of problems encountered in graduate research.

1 Guidelines

1. System Selection:
   - Choose a classical mechanical system. Options include but are not limited to the double pendulum, the three-body problem, non-linear dynamics, chaotic systems, systems with random perturbations, and systems with bifurcations.
   - Students should work with Dr. Jay to find a suitable system and define a reasonable scope, either during office hours, after class, or by appointment.
2. Literature Review:
   - The project is *not* expected to involve producing novel scientific results—that's the end goal of the graduate program!
   - After identifying a system to study, you should conduct a brief literature review to answer the following questions:
     - What is the physical relevance of the system?
     - What are the key equations (e.g., the Lagrangian, Hamiltonian, or equations of motion)?
     - What is known about the system and its solutions, and how are such solutions typically obtained?
     - What methods are appropriate to study this system?
3. Analytical and Numerical Analysis:
   - Develop a numerical solution to the system in python, applying appropriate methods (e.g., Runge-Kutta or symplectic integrators).
   - Perform an analytical analysis where possible (e.g., in limiting cases where the system is integrable).
   - Ensure your numerical analysis includes error analysis and stability considerations to validate your results.
   - Include relevant diagrams of the system's trajectory, phase curves, or phase flow as relevant.

2 Deliverables

1. **Scientific Narrative (3-5 pages)**: You should provide a scientific narrative summarizing your results. The target audience should be beginning graduate students who have just completed PH621.

   The scientific narrative should be a clear and concise document, typed using standard word processing software (e.g., "Word" or LaTeX). The document should use standard fonts (e.g., 12-point Times New Roman or Computer Modern) with double spacing and 1-inch margins. The essential points are readability and clarity of presentation, not some particular font or margin size.

   The narrative will include a descriptive title, a brief abstract (150-250 words), and sections such as Introduction, Methodology, and Results. Use formal scientific language and maintain an objective tone, avoiding colloquialisms. Present your findings clearly with appropriate figures or tables. Ensure technical terms are defined. Include a bibliography formatted in a consistent citation style. Proofread for grammatical correctness and coherence throughout. Aim for clarity and tight logical flow to convey effectively your findings and results.

2. **Python code**: You should submit well-documented Python code that correctly implements the numerical methods used in your analysis. The code should be organized, with comments explaining key sections and functions. The code should be able to run on another computer to reproduce all the results (e.g., tables or figures) presented in the scientific narrative.

3. **Oral Presentation (10 minutes)**: Prepare a concise presentation using standard software (e.g., Powerpoint, Keynote, or LaTeX Beamer) to share your findings with the class. Focus on key insights, your methodology, and the relevance of your work to classical mechanics. Remember that oral presentations are about communicating a scientific story. Every talk is a job talk!

3 Evaluation Criteria

- Depth of analysis and understanding of the chosen system.
- Integration and effectiveness of appropriate numerical methods.
- Clarity and organization of the scientific narrative.
- Quality, clarity, correctness of the Python code.
- Effectiveness of the oral presentation and engagement with the audience.

4 Timeline

(a) System selection/proposal due: **Friday, October 23, 2026**.

(b) One-on-one meeting to discuss code progress: **Friday, November 6, 2025**.

(c) Draft narrative due: **Friday, November 20, 2026**.

(d) Final deliverables due: **Last week of classes**.